



Case pattern

Re-run a chain of features for every case of a Case template -- each case with its own selections and values -- and name the new bodies by case.



What it does

'Apply to each' for the feature tree. Onshape's Pattern moves copies by a transform; Evan Reese's Query Pattern rebinds one seed. Case pattern rebinds up to 8 query variables and 4 values per case and re-runs the listed features for each case.

Use it when

- A detail repeats on places that differ in shape
- Each place needs its own dimensions
- The chain edits existing geometry (fillets, move face) as well as making new

T1: one chain (Boss, rim fillet, boss-edge fillet) built on case A, replayed for the other two blocks (side view)

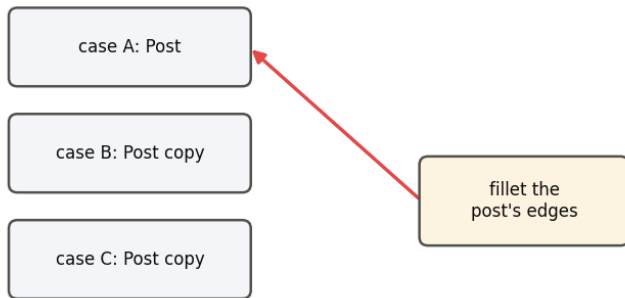


T1: the boss chain on three blocks, each at its own height and radius.

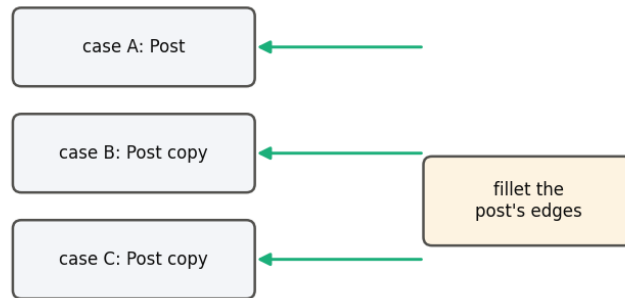


The one rule

Clicked (or qCreatedBy(makeld(...)))
the fillet stays on case A's edges
-> test T3 / T4: ERROR



Query variable 'created by Post'
each case's copy is remapped
-> test T5: every post filleted



Inside the list

Geometry one listed feature made, used by another, must come through a Query variable 'created by' that feature.

Why

Each case runs in a pattern frame that remaps those references onto the case's copies; clicks stay on case 1.

Outside the list

References to existing geometry are fine: they are rebound through the template's names.



The dialog

T1 Case pattern -> Boss_...  

1 Case template
T1 Case template A: ?, ?; valu... x

2 Features to repeat
T1 Boss: extrude ? ? new x
T1 Rim: fillet ? 2 mm x
T1 ? = edges created by Boss ... x
T1 Boss edges: fillet 2.2 mm x

3 > Keep

4 Name separator _
 

#	Parameter	What it means
1	Case template	Exactly one Case template.
2	Features to repeat	The chain built on the template's names.
3	Keep	Which new bodies to keep per case (sketches off by default).
4	Name separator	Between name and case (default _): Boss_A -> Boss_B.



Example: bosses on three blocks (T1)

T1: one chain (Boss, rim fillet, boss-edge fillet) built on case A, replayed for the other two blocks (side view)



case A (template): #bossH 15



case B: #bossH 25, R3



case C: #bossH 8, R2

Chain: Boss (extrude #top by #bossH), rim fillet 2 mm, #bossEdges = edges created by Boss, fillet #bossEdges at #edgeR

Case A: 15 mm, R3 (the template). Case B: 25 mm, R3. Case C (pentagon): 8 mm, R2. New bodies are named Boss_<case>. (In the studio case B is currently named 'Foobar!'.)



Outputs and messages

Published for Extract variables

Key	What it holds
output	The bodies each case created (filtered by Keep), named <name>_<case>.

Messages

	Message / what to do
Warning	Not built: B: input 1 (#top) selects nothing; C: feature 2 failed (...) The other cases are still built.
Error	No case was built.
Info	N bodies kept Onshape's default name Edit the Case pattern once to refresh case 1's names.



Tips and limits



Sketches

Re-solved per case, but references to input geometry stay on case 1. Build sketches on geometry derived from the inputs.



Order matters

Cases run in order and see earlier results; a failed case cannot undo edits to existing geometry.



Cost

(cases + 1) x the chain on every rebuild.



Verified

Fillet and Move face on existing geometry, new extrudes, native query variables (T1-T5).



Where it lives

Document	Case_Pattern (2099413dd91f34578b385892)
Tab	case_pattern (both features)
Tests	Case pattern tests studio: T1-T5 (T3, T4 must error) - devtools/onshape/build_case_pattern_tests.py
Explainer	docs/explainers/case_pattern/case_pattern_explained.md
Related	Case template; Query variable